

CoRoT-2b

R-2676

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_260726 · created 2026-08-30T23:00:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461247.8784 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.087 +/- 0.065
Transit depth (Rp/Rs)^2	0.76 +/- 1.13 [%]
Orbital Inclination (inc)	87.36 +/- 2.08
Airmass coefficient 1 (a1)	1.068 +/- 0.094
Airmass coefficient 2 (a2)	-0.025 +/- 0.087
Scatter in the residuals of the lightcurve fit is	8.78 %
Best Comparison Star	#3 - [547, 317]
Optimal Method	PSF photometry
Transit Duration (day)	0.084 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.087 ± 0.065 vs 0.1667 (deviation 1.23σ)
Duration fit vs published	0.084 d vs 0.095 d (deviation 1.16σ)
Mid-time O-C	11.5 min
Depth SNR	1.3
Residual scatter	8.78 %

Light curve

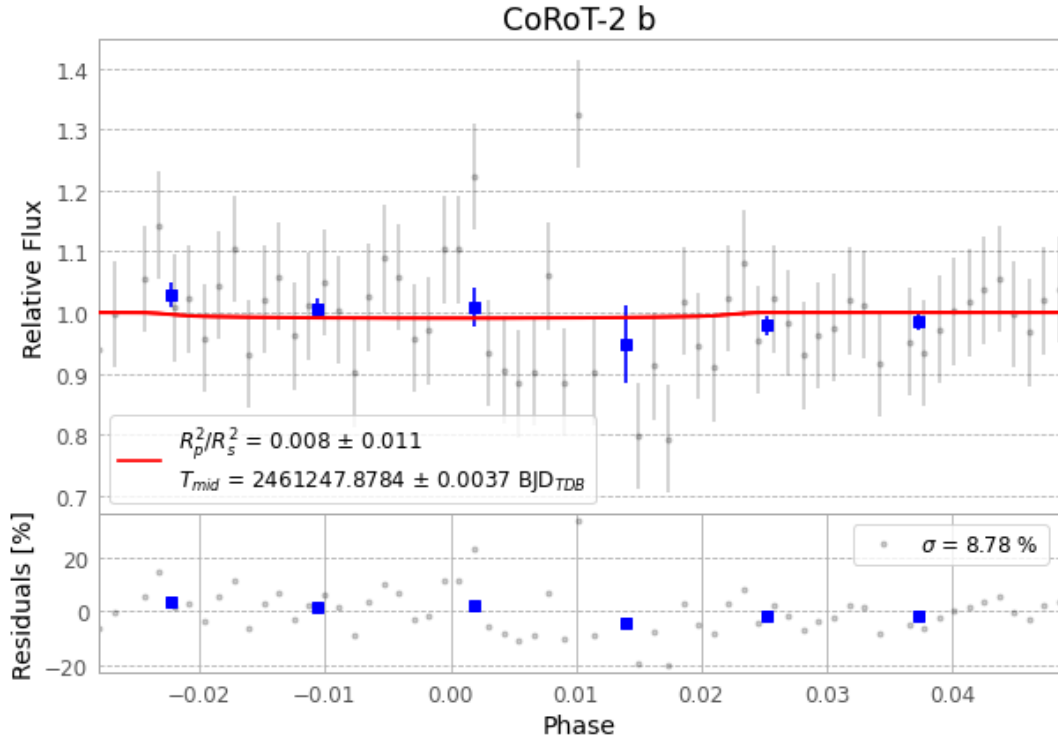


Figure 1 — FinalLightCurve_CoRoT-2 b_2026-07-26.png (SHA-256 9566fa8535df11c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 74], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e47e41141bfbbc97...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), CoRoT-2260726074515.FITS → CoRoT-2260726105717.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-260726.log (4088fee428c7...), FinalLightCurve_CoRoT-2 b_2026-07-26.png (9566fa8535df...), FinalLightCurve_CoRoT-2 b_2026-07-26.csv (819ab5adf589...)

Compute resources

Wall time 272.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 23:00Z.